Chapter 2

Mathematical Tools of Quantum Mechanics

2.1 Introduction

We deal here with the mathematical machinery needed to study quantum mechanics. Although this chapter is mathematical in scope, no attempt is made to be mathematically complete or rigorous. We limit ourselves to those practical issues that are relevant to the formalism of quantum mechanics.

The Schrödinger equation is one of the cornerstones of the theory of quantum mechanics; it has the structure of a *linear* equation. The formalism of quantum mechanics deals with operators that are linear and wave functions that belong to an abstract Hilbert space. The mathematical properties and structure of Hilbert spaces are essential for a proper understanding of the formalism of quantum mechanics. For this, we are going to review briefly the properties of Hilbert spaces and those of linear operators. We will then consider Dirac's *bra-ket* notation.

Quantum mechanics was formulated in two different ways by Schrödinger and Heisenberg. Schrödinger's wave mechanics and Heisenberg's matrix mechanics are the representations of the general formalism of quantum mechanics in *continuous* and *discrete* basis systems, respectively. For this, we will also examine the mathematics involved in representing kets, bras, bra-kets, and operators in discrete and continuous bases.

2.2 The Hilbert Space and Wave Functions

2.2.1 The Linear Vector Space

A linear vector space consists of two sets of elements and two algebraic rules:

- a set of *vectors* ψ, ϕ, χ, ... and a set of *scalars* $a, b, c, \ldots$;
- a rule for vector *addition* and a rule for scalar *multiplication*.

(a) Addition rule
The addition rule has the properties and structure of an *abelian* group:

- If ψ and ϕ are vectors (elements) of a space, their sum, $\psi + \phi$, is also a vector of the same space.
- Commutativity: $\psi + \phi = \phi + \psi$.
- Associativity: $(\psi + \phi) + \chi = \psi + (\phi + \chi)$.
- Existence of a zero or neutral vector: for each vector ψ, there must exist a zero vector O such that: $O + \psi = \psi + O = \psi$.
- Existence of a symmetric or inverse vector: each vector ψ must have a symmetric vector $(-\psi)$ such that $\psi + (-\psi) = (-\psi) + \psi = O$.

(b) Multiplication rule
The multiplication of vectors by scalars (scalars can be real or complex numbers) has these properties:

- The product of a scalar with a vector gives another vector. In general, if ψ and ϕ are two vectors of the space, any linear combination $a\psi + b\phi$ is also a vector of the space, a and b being scalars.
- Distributivity with respect to addition:

$$a(\psi + \phi) = a\psi + a\phi, \qquad (a + b)\psi = a\psi + b\psi, \tag{2.1}$$

- Associativity with respect to multiplication of scalars:

$$a(b\psi) = (ab)\psi \tag{2.2}$$

- For each element ψ there must exist a unitary scalar I and a zero scalar "o" such that

$$I\psi = \psi I = \psi \quad \text{and} \quad o\psi = \psi o = o. \tag{2.3}$$

2.2.2 The Hilbert Space

A Hilbert space $\mathcal{H}$ consists of a set of vectors $\psi, \phi, \chi, \ldots$ and a set of *scalars* $a, b, c, \ldots$ which satisfy the following *four* properties:

(a) $\mathcal{H}$ is a linear space

The properties of a linear space were considered in the previous section.

(b) $\mathcal{H}$ has a defined scalar product that is strictly positive

The scalar product of an element ψ with another element ϕ is in general a complex number, denoted by (ψ, ϕ), where $(\psi, \phi) =$ complex number. **Note:** Watch out for the order! Since the scalar product is a complex number, the quantity (ψ, ϕ) is generally not equal to (ϕ, ψ): $(\psi, \phi) = \psi^* \phi$ while $(\phi, \psi) = \phi^* \psi$. The scalar product satisfies the following properties:

- The scalar product of ψ with ϕ is equal to the complex conjugate of the scalar product of ϕ with ψ:

$$(\psi, \phi) = (\phi, \psi)^*. \tag{2.4}$$

- The scalar product of ϕ with ψ is linear with respect to the second factor if $\psi = a\psi_1 + b\psi_2$:

$$(\phi, a\psi_1 + b\psi_2) = a(\phi, \psi_1) + b(\phi, \psi_2), \tag{2.5}$$

and antilinear with respect to the first factor if $\phi = a\phi_1 + b\phi_2$:

$$(a\phi_1 + b\phi_2, \psi) = a^*(\phi_1, \psi) + b^*(\phi_2, \psi). \tag{2.6}$$

- The scalar product of a vector ψ with itself is a positive real number:

$$(\psi, \psi) = \| \psi \|^2 \geq 0, \tag{2.7}$$

where the equality holds only for $\psi = O$.

(c) $\mathcal{H}$ is separable

There exists a Cauchy sequence $\psi_n \in \mathcal{H}$ $(n = 1, 2, \ldots)$ such that for every ψ of $\mathcal{H}$ and $\varepsilon > 0$, there exists at least one ψ_n of the sequence for which

$$\| \psi - \psi_n \| < \varepsilon. \tag{2.8}$$

(d) $\mathcal{H}$ is complete

Every Cauchy sequence $\psi_n \in \mathcal{H}$ converges to an element of $\mathcal{H}$. That is, for any ψ_n, the relation

$$\lim_{n,m\to\infty} \| \psi_n - \psi_m \| = 0, \tag{2.9}$$

defines a unique limit ψ of $\mathcal{H}$ such that

$$\lim_{n\to\infty} \| \psi - \psi_n \| = 0. \tag{2.10}$$

Remark
We should note that in a scalar product (ϕ, ψ), the second factor, ψ, belongs to the Hilbert space $\mathcal{H}$, while the first factor, ϕ, belongs to its dual Hilbert space $\mathcal{H}_d$. The distinction between $\mathcal{H}$ and $\mathcal{H}_d$ is due to the fact that, as mentioned above, the scalar product is not commutative: $(\phi, \psi) \neq (\psi, \phi)$; the order matters! From linear algebra, we know that every vector space can be associated with a dual vector space.

2.2.3 Dimension and Basis of a Vector Space

A set of N nonzero vectors $\phi_1, \phi_2, \ldots, \phi_N$ is said to be *linearly independent* if and only if the solution of the equation

$$\sum_{i=1}^{N} a_i \phi_i = 0 \tag{2.11}$$

is $a_1 = a_2 = \cdots = a_N = 0$. But if there exists a set of scalars, which are not all zero, so that one of the vectors (say ϕ_n) can be expressed as a linear combination of the others,

$$\phi_n = \sum_{i=1}^{n-1} a_i \phi_i + \sum_{i=n+1}^{N} a_i \phi_i, \tag{2.12}$$

the set $\{\phi_i\}$ is said to be *linearly dependent*.

Dimension: The *dimension* of a vector space is given by the *maximum number* of linearly independent vectors the space can have. For instance, if the maximum number of linearly independent vectors a space has is N (i.e., $\phi_1, \phi_2, \ldots, \phi_N$), this space is said to be N-dimensional. In this N-dimensional vector space, any vector ψ can be expanded as a linear combination:

$$\psi = \sum_{i=1}^{N} a_i \phi_i. \tag{2.13}$$

Basis: The *basis* of a vector space consists of a set of the maximum possible number of linearly independent vectors belonging to that space. This set of vectors, $\phi_1, \phi_2, \ldots, \phi_N$, to be denoted in short by $\{\phi_i\}$, is called the basis of the vector space, while the vectors $\phi_1, \phi_2, \ldots, \phi_N$ are called the base vectors. Although the set of these linearly independent vectors is arbitrary, it is convenient to choose them *orthonormal*; that is, their scalar products satisfy the relation $(\phi_i, \phi_j) = \delta_{ij}$ (we may recall that $\delta_{ij} = 1$ whenever $i = j$ and zero otherwise). The basis is said to be *orthonormal* if it consists of a set of orthonormal vectors. Moreover, the basis is said to be *complete* if it spans the entire space; that is, there is no need to introduce any additional base vector. The expansion coefficients a_i in (2.13) are called the *components* of the vector ψ in the basis. Each component is given by the scalar product of ψ with the corresponding base vector, $a_j = (\phi_j, \psi)$.

Examples of linear vector spaces

Let us give two examples of linear spaces that are Hilbert spaces: one having a *finite (discrete)* set of base vectors, the other an *infinite (continuous)* basis.

- The first one is the three-dimensional Euclidean vector space; the basis of this space consists of three linearly independent vectors, usually denoted by $\vec{i}, \vec{j}, \vec{k}$. Any vector of the Euclidean space can be written in terms of the base vectors as $\vec{A} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, where a_1, a_2, and a_3 are the components of $\vec{A}$ in the basis; each component can be determined by taking the scalar product of $\vec{A}$ with the corresponding base vector: $a_1 = \vec{i} \cdot \vec{A}$, $a_2 = \vec{j} \cdot \vec{A}$, and $a_3 = \vec{k} \cdot \vec{A}$. Note that the scalar product in the Euclidean space is real and hence symmetric. The norm in this space is the usual length of vectors $\| \vec{A} \| = A$. Note also that whenever $a_1\vec{i} + a_2\vec{j} + a_3\vec{k} = 0$ we have $a_1 = a_2 = a_3 = 0$ and that none of the unit vectors $\vec{i}, \vec{j}, \vec{k}$ can be expressed as a linear combination of the other two.

- The second example is the space of the entire complex functions $\psi(x)$; the dimension of this space is infinite for it has an infinite number of linearly independent basis vectors.

Example 2.1

Check whether the following sets of functions are linearly independent or dependent on the real x-axis.

(a) $f(x) = 4, g(x) = x^2, h(x) = e^{2x}$
(b) $f(x) = x, g(x) = x^2, h(x) = x^3$
(c) $f(x) = x, g(x) = 5x, h(x) = x^2$
(d) $f(x) = 2 + x^2, g(x) = 3 - x + 4x^3, h(x) = 2x + 3x^2 - 8x^3$

Solution

(a) The first set is clearly linearly independent since $a_1 f(x) + a_2 g(x) + a_3 h(x) = 4a_1 + a_2 x^2 + a_3 e^{2x} = 0$ implies that $a_1 = a_2 = a_3 = 0$ for any value of x.

(b) The functions $f(x) = x$, $g(x) = x^2$, $h(x) = x^3$ are also linearly independent since $a_1 x + a_2 x^2 + a_3 x^3 = 0$ implies that $a_1 = a_2 = a_3 = 0$ no matter what the value of x. For instance, taking $x = -1, 1, 3$, the following system of three equations

$$-a_1 + a_2 - a_3 = 0, \qquad a_1 + a_2 + a_3 = 0, \qquad 3a_1 + 9a_2 + 27a_3 = 0 \tag{2.14}$$

yields $a_1 = a_2 = a_3 = 0$.

(c) The functions $f(x) = x$, $g(x) = 5x$, $h(x) = x^2$ are not linearly independent, since $g(x) = 5f(x) + 0 \times h(x)$.

(d) The functions $f(x) = 2 + x^2$, $g(x) = 3 - x + 4x^3$, $h(x) = 2x + 3x^2 - 8x^3$ are not linearly independent since $h(x) = 3f(x) - 2g(x)$.

Example 2.2

Are the following sets of vectors (in the three-dimensional Euclidean space) linearly independent or dependent?

(a) $\vec{A} = (3, 0, 0)$, $\vec{B} = (0, -2, 0)$, $\vec{C} = (0, 0, -1)$

(b) $\vec{A} = (6, -9, 0)$, $\vec{B} = (-2, 3, 0)$

(c) $\vec{A} = (2, 3, -1)$, $\vec{B} = (0, 1, 2)$, $\vec{C} = (0, 0, -5)$

(d) $\vec{A} = (1, -2, 3)$, $\vec{B} = (-4, 1, 7)$, $\vec{C} = (0, 10, 11)$, and $\vec{D} = (14, 3, -4)$

Solution

(a) The three vectors $\vec{A} = (3, 0, 0)$, $\vec{B} = (0, -2, 0)$, $\vec{C} = (0, 0, -1)$ are linearly independent, since

$$a_1\vec{A} + a_2\vec{B} + a_3\vec{C} = 0 \Longrightarrow 3a_1\vec{i} - 2a_2\vec{j} - a_3\vec{k} = 0 \tag{2.15}$$

leads to

$$3a_1 = 0, \qquad -2a_2 = 0, \qquad -a_3 = 0, \tag{2.16}$$

which yields $a_1 = a_2 = a_3 = 0$.

(b) The vectors $\vec{A} = (6, -9, 0)$, $\vec{B} = (-2, 3, 0)$ are linearly dependent, since the solution to

$$a_1\vec{A} + a_2\vec{B} = 0 \quad \Longrightarrow \quad (6a_1 - 2a_2)\vec{i} + (-9a_1 + 3a_2)\vec{j} = 0 \tag{2.17}$$

is $a_1 = a_2/3$. The first vector is equal to -3 times the second one: $\vec{A} = -3\vec{B}$.

(c) The vectors $\vec{A} = (2, 3, -1)$, $\vec{B} = (0, 1, 2)$, $\vec{C} = (0, 0, -5)$ are linearly independent, since

$$a_1\vec{A} + a_2\vec{B} + a_3\vec{C} = 0 \Longrightarrow 2a_1\vec{i} + (3a_1 + a_2)\vec{j} + (-a_1 + 2a_2 - 5a_3)\vec{k} = 0 \tag{2.18}$$

leads to

$$2a_1 = 0, \qquad 3a_1 + a_2 = 0, \qquad -a_1 + 2a_2 - 5a_3 = 0. \tag{2.19}$$

The only solution of this system is $a_1 = a_2 = a_3 = 0$.

(d) The vectors $\vec{A} = (1, -2, 3)$, $\vec{B} = (-4, 1, 7)$, $\vec{C} = (0, 10, 11)$, and $\vec{D} = (14, 3, -4)$ are not linearly independent, because $\vec{D}$ can be expressed in terms of the other vectors:

$$\vec{D} = 2\vec{A} - 3\vec{B} + \vec{C}. \tag{2.20}$$

2.2.4 Square-Integrable Functions: Wave Functions

In the case of function spaces, a "vector" element is given by a *complex function* and the *scalar product* by *integrals*. That is, the scalar product of two functions $\psi(x)$ and $\phi(x)$ is given by

$$(\psi, \phi) = \int \psi^*(x)\phi(x)\, dx. \tag{2.21}$$

If this integral *diverges*, the scalar product *does not exist*. As a result, if we want the function space to possess a scalar product, we must select only those functions for which (ψ, ϕ) is *finite*. In particular, a function $\psi(x)$ is said to be *square integrable* if the scalar product of ψ with itself,

$$(\psi, \psi) = \int |\psi(x)|^2\, dx, \tag{2.22}$$

is *finite*.

It is easy to verify that the space of square-integrable functions possesses the properties of a Hilbert space. For instance, any linear combination of square-integrable functions is also a square-integrable function and (2.21) satisfies all the properties of the scalar product of a Hilbert space.

Note that the dimension of the Hilbert space of square-integrable functions is infinite, since each wave function can be expanded in terms of an infinite number of linearly independent functions. The dimension of a space is given by the maximum number of linearly independent basis vectors required to span that space.

A good example of square-integrable functions is the *wave function* of quantum mechanics, $\psi(\vec{r}, t)$. We have seen in Chapter 1 that, according to Born's probabilistic interpretation of $\psi(\vec{r}, t)$, the quantity $|\ \psi(\vec{r}, t)\ |^2\ d^3r$ represents the probability of finding, at time t, the particle in a volume d^3r, centered around the point $\vec{r}$. The probability of finding the particle somewhere in space must then be equal to 1:

$$\int |\ \psi(\vec{r}, t)\ |^2\ d^3r = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy \int_{-\infty}^{+\infty} |\ \psi(\vec{r}, t)\ |^2\ dz = 1; \tag{2.23}$$

hence the wave functions of quantum mechanics are square-integrable. Wave functions satisfying (2.23) are said to be normalized or square-integrable. As wave mechanics deals with square-integrable functions, any wave function which is not square-integrable has no physical meaning in quantum mechanics.

2.3 Dirac Notation

The physical state of a system is represented in quantum mechanics by elements of a Hilbert space; these elements are called state vectors. We can represent the state vectors in different bases by means of function expansions. This is analogous to specifying an ordinary (Euclidean) vector by its components in various coordinate systems. For instance, we can represent equivalently a vector by its components in a Cartesian coordinate system, in a spherical coordinate system, or in a cylindrical coordinate system. *The meaning of a vector is, of course, independent of the coordinate system chosen to represent its components*. Similarly, the state of a microscopic system has a meaning independent of the basis in which it is expanded.

To free state vectors from coordinate meaning, Dirac introduced what was to become an invaluable notation in quantum mechanics; it allows one to manipulate the formalism of quantum

mechanics with ease and clarity. He introduced the concepts of kets, bras, and bra-kets, which will be explained below.

Kets: elements of a vector space
Dirac denoted the state vector ψ by the symbol $|\ \psi\rangle$, which he called a *ket* vector, or simply a ket. Kets belong to the Hilbert (vector) space $\mathcal{H}$, or, in short, to the ket-space.

Bras: elements of a dual space
As mentioned above, we know from linear algebra that a dual space can be associated with every vector space. Dirac denoted the elements of a dual space by the symbol $\langle\ |$, which he called a bra vector, or simply a bra; for instance, the element $\langle\psi\ |$ represents a bra. *Note: For every ket* $|\ \psi\rangle$ *there exists a unique bra* $\langle\psi\ |$ *and vice versa*. Again, while kets belong to the Hilbert space $\mathcal{H}$, the corresponding bras belong to its dual (Hilbert) space $\mathcal{H}_d$.

Bra-ket: Dirac notation for the scalar product
Dirac denoted the scalar (inner) product by the symbol $\langle\ |\ \rangle$, which he called a a *bra-ket*. For instance, the scalar product (ϕ, ψ) is denoted by the bra-ket $\langle\phi\ |\ \psi\rangle$:

$$(\phi, \psi) \quad \longrightarrow \quad \langle\phi\ |\ \psi\rangle. \tag{2.24}$$

Note: When a ket (or bra) is multiplied by a complex number, we also get a ket (or bra).

Remark: In wave mechanics we deal with wave functions $\psi(\vec{r}, t)$, but in the more general formalism of quantum mechanics we deal with abstract kets $|\ \psi\rangle$. Wave functions, like kets, are elements of a Hilbert space. We should note that, like a wave function, a ket represents the system completely, and hence knowing $|\ \psi\rangle$ means knowing all its amplitudes in all possible representations. As mentioned above, kets are independent of any particular representation. There is no reason to single out a particular representation basis such as the representation in the position space. Of course, if we want to know the probability of finding the particle at some position in space, we need to work out the formalism within the coordinate representation. The state vector of this particle at time t will be given by the spatial wave function $\langle\vec{r}, t\ |\ \psi\rangle = \psi(\vec{r}, t)$. In the coordinate representation, the scalar product $\langle\phi\ |\ \psi\rangle$ is given by

$$\langle\phi\ |\ \psi\rangle = \int \phi^*(\vec{r}, t)\,\psi(\vec{r}, t)\,d^3r. \tag{2.25}$$

Similarly, if we are considering the three-dimensional momentum of a particle, the ket $|\ \psi\rangle$ will have to be expressed in momentum space. In this case the state of the particle will be described by a wave function $\psi(\vec{p}, t)$, where $\vec{p}$ is the momentum of the particle.

Properties of kets, bras, and bra-kets

- **Every ket has a corresponding bra**

  To every *ket* $|\ \psi\rangle$, there corresponds a unique *bra* $\langle\psi\ |$ and vice versa:

  $$|\ \psi\rangle \quad \longleftrightarrow \quad \langle\psi\ |\ . \tag{2.26}$$

  There is a one-to-one correspondence between bras and kets:

  $$a\ |\ \psi\rangle + b\ |\ \phi\rangle \quad \longleftrightarrow \quad a^*\langle\psi\ | + b^*\langle\phi\ |, \tag{2.27}$$

  where a and b are complex numbers. The following is a common notation:

  $$|\ a\psi\rangle = a\ |\ \psi\rangle, \qquad \langle a\psi\ | = a^*\langle\psi\ |\ . \tag{2.28}$$

- **Properties of the scalar product**

  In quantum mechanics, since the scalar product is a complex number, the ordering matters a lot. We must be careful to distinguish a scalar product from its complex conjugate; $\langle \psi \mid \phi \rangle$ is not the same thing as $\langle \phi \mid \psi \rangle$:

$$\langle \phi \mid \psi \rangle^* = \langle \psi \mid \phi \rangle. \tag{2.29}$$

  This property becomes clearer if we apply it to (2.21):

$$\langle \phi \mid \psi \rangle^* = \left(\int \phi^*(\vec{r}, t) \psi(\vec{r}, t)\, d^3r \right)^* = \int \psi^*(\vec{r}, t) \phi(\vec{r}, t)\, d^3r = \langle \psi \mid \phi \rangle. \tag{2.30}$$

  When $\mid \psi \rangle$ and $\mid \phi \rangle$ are real, we would have $\langle \psi \mid \phi \rangle = \langle \phi \mid \psi \rangle$. Let us list some additional properties of the scalar product:

$$\langle \psi \mid a_1 \psi_1 + a_2 \psi_2 \rangle = a_1 \langle \psi \mid \psi_1 \rangle + a_2 \langle \psi \mid \psi_2 \rangle, \tag{2.31}$$

$$\langle a_1 \phi_1 + a_2 \phi_2 \mid \psi \rangle = a_1^* \langle \phi_1 \mid \psi \rangle + a_2^* \langle \phi_2 \mid \psi \rangle, \tag{2.32}$$

$$\begin{aligned} \langle a_1 \phi_1 + a_2 \phi_2 \mid b_1 \psi_1 + b_2 \psi_2 \rangle = {} & a_1^* b_1 \langle \phi_1 \mid \psi_1 \rangle + a_1^* b_2 \langle \phi_1 \mid \psi_2 \rangle \\ & + a_2^* b_1 \langle \phi_2 \mid \psi_1 \rangle + a_2^* b_2 \langle \phi_2 \mid \psi_2 \rangle. \end{aligned} \tag{2.33}$$

- **The norm is real and positive**

  For any state vector $\mid \psi \rangle$ of the Hilbert space $\mathcal{H}$, the norm $\langle \psi \mid \psi \rangle$ is real and positive; $\langle \psi \mid \psi \rangle$ is equal to zero only for the case where $\mid \psi \rangle = O$, where O is the zero vector. If the state $\mid \psi \rangle$ is normalized then $\langle \psi \mid \psi \rangle = 1$.

- **Schwarz inequality**

  For any two states $\mid \psi \rangle$ and $\mid \phi \rangle$ of the Hilbert space, we can show that

$$|\langle \psi \mid \phi \rangle|^2 \leq \langle \psi \mid \psi \rangle \langle \phi \mid \phi \rangle. \tag{2.34}$$

  If $\mid \psi \rangle$ and $\mid \phi \rangle$ are linearly dependent (i.e., proportional: $\mid \psi \rangle = \alpha \mid \phi \rangle$, where α is a scalar), this relation becomes an equality. The Schwarz inequality (2.34) is analogous to the following relation of the real Euclidean space

$$|\vec{A} \cdot \vec{B}|^2 \leq |\vec{A}|^2 \, |\vec{B}|^2 . \tag{2.35}$$

- **Triangle inequality**

$$\sqrt{\langle \psi + \phi \mid \psi + \phi \rangle} \leq \sqrt{\langle \psi \mid \psi \rangle} + \sqrt{\langle \phi \mid \phi \rangle}. \tag{2.36}$$

  If $\mid \psi \rangle$ and $\mid \phi \rangle$ are linearly dependent, $\mid \psi \rangle = \alpha \mid \phi \rangle$, and if the proportionality scalar α is real and positive, the triangle inequality becomes an equality. The counterpart of this inequality in Euclidean space is given by $|\vec{A} + \vec{B}| \leq |\vec{A}| + |\vec{B}|$.

- **Orthogonal states**

  Two *kets*, $\mid \psi \rangle$ and $\mid \phi \rangle$, are said to be orthogonal if they have a vanishing scalar product:

$$\langle \psi \mid \phi \rangle = 0. \tag{2.37}$$

- **Orthonormal states**

  Two *kets*, $|\ \psi\rangle$ and $|\ \phi\rangle$, are said to be orthonormal if they are orthogonal and if each one of them has a unit norm:

$$\langle\psi\ |\ \phi\rangle = 0, \qquad \langle\psi\ |\ \psi\rangle = 1, \qquad \langle\phi\ |\ \phi\rangle = 1. \tag{2.38}$$

- **Forbidden quantities**

  If $|\ \psi\rangle$ and $|\ \phi\rangle$ belong to the same vector (Hilbert) space, products of the type $|\ \psi\rangle\ |\ \phi\rangle$ and $\langle\psi\ |\ \langle\phi\ |$ are forbidden. They are nonsensical, since $|\ \psi\rangle\ |\ \phi\rangle$ and $\langle\psi\ |\ \langle\phi\ |$ are neither kets nor bras (an explicit illustration of this will be carried out in the example below and later on when we discuss the representation in a discrete basis). If $|\ \psi\rangle$ and $|\ \phi\rangle$ belong, however, to different vector spaces (e.g., $|\ \psi\rangle$ belongs to a spin space and $|\ \phi\rangle$ to an orbital angular momentum space), then the product $|\ \psi\rangle\ |\ \phi\rangle$, written as $|\ \psi\rangle \otimes |\ \phi\rangle$, represents a tensor product of $|\ \psi\rangle$ and $|\ \phi\rangle$. Only in these typical cases are such products meaningful.

Example 2.3

(**Note:** We will see later in this chapter that kets are represented by column matrices and bras by row matrices; this example is offered earlier than it should because we need to show some concrete illustrations of the formalism.) Consider the following two kets:

$$|\ \psi\rangle = \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix}, \qquad |\ \phi\rangle = \begin{pmatrix} 2 \\ -i \\ 2-3i \end{pmatrix}.$$

(a) Find the bra $\langle\phi\ |$.
(b) Evaluate the scalar product $\langle\phi\ |\ \psi\rangle$.
(c) Examine why the products $|\ \psi\rangle\ |\ \phi\rangle$ and $\langle\phi\ |\ \langle\psi\ |$ do not make sense.

Solution

(a) As will be explained later when we introduce the Hermitian adjoint of kets and bras, we want to mention that the bra $\langle\phi\ |$ can be obtained by simply taking the complex conjugate of the transpose of the ket $|\ \phi\rangle$:

$$\langle\phi\ | = (2 \quad i \quad 2+3i)\,. \tag{2.39}$$

(b) The scalar product $\langle\phi\ |\ \psi\rangle$ can be calculated as follows:

$$\begin{aligned} \langle\phi\ |\ \psi\rangle &= (2 \quad i \quad 2+3i) \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix} \\ &= 2(-3i) + i(2+i) + 4(2+3i) \\ &= 7 + 8i. \end{aligned} \tag{2.40}$$

(c) First, the product $|\ \psi\rangle\ |\ \phi\rangle$ cannot be performed because, from linear algebra, the product of two column matrices cannot be performed. Similarly, since two row matrices cannot be multiplied, the product $\langle\phi\ |\ \langle\psi\ |$ is meaningless.

Physical meaning of the scalar product

The scalar product can be interpreted in two ways. First, by analogy with the scalar product of ordinary vectors in the Euclidean space, where $\vec{A} \cdot \vec{B}$ represents the projection of $\vec{B}$ on $\vec{A}$, the product $\langle \phi \mid \psi \rangle$ also represents the projection of $\mid \psi \rangle$ onto $\mid \phi \rangle$. Second, in the case of normalized states and according to Born's probabilistic interpretation, the quantity $\langle \phi \mid \psi \rangle$ represents the probability amplitude that the system's state $\mid \psi \rangle$ will, after a measurement is performed on the system, be found to be in another state $\mid \phi \rangle$.

Example 2.4 (Bra-ket algebra)

Consider the states $\mid \psi \rangle = 3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle$ and $\mid \chi \rangle = - \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle$, where $\mid \phi_1 \rangle$ and $\mid \phi_2 \rangle$ are orthonormal.

(a) Calculate $\mid \psi + \chi \rangle$ and $\langle \psi + \chi \mid$.

(b) Calculate the scalar products $\langle \psi \mid \chi \rangle$ and $\langle \chi \mid \psi \rangle$. Are they equal?

(c) Show that the states $\mid \psi \rangle$ and $\mid \chi \rangle$ satisfy the Schwarz inequality.

(d) Show that the states $\mid \psi \rangle$ and $\mid \chi \rangle$ satisfy the triangle inequality.

Solution

(a) The calculation of $\mid \psi + \chi \rangle$ is straightforward:

$$\begin{aligned} \mid \psi + \chi \rangle &= \mid \psi \rangle + \mid \chi \rangle = (3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle) + (- \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle) \\ &= (-1 + 3i) \mid \phi_1 \rangle - 5i \mid \phi_2 \rangle. \end{aligned} \tag{2.41}$$

This leads at once to the expression of $\langle \psi + \chi \mid$:

$$\langle \psi + \chi \mid = (-1 + 3i)^* \langle \phi_1 \mid + (-5i)^* \langle \phi_2 \mid = (-1 - 3i) \langle \phi_1 \mid + 5i \langle \phi_2 \mid . \tag{2.42}$$

(b) Since $\langle \phi_1 \mid \phi_1 \rangle = \langle \phi_2 \mid \phi_2 \rangle = 1$, $\langle \phi_1 \mid \phi_2 \rangle = \langle \phi_2 \mid \phi_1 \rangle = 0$, and since the bras corresponding to the kets $\mid \psi \rangle = 3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle$ and $\mid \chi \rangle = - \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle$ are given by $\langle \psi \mid = -3i \langle \phi_1 \mid + 7i \langle \phi_2 \mid$ and $\langle \chi \mid = - \langle \phi_1 \mid - 2i \langle \phi_2 \mid$, the scalar products are

$$\begin{aligned} \langle \psi \mid \chi \rangle &= (-3i \langle \phi_1 \mid + 7i \langle \phi_2 \mid) (- \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle) \\ &= (-3i)(-1) \langle \phi_1 \mid \phi_1 \rangle + (7i)(2i) \langle \phi_2 \mid \phi_2 \rangle \\ &= -14 + 3i, \end{aligned} \tag{2.43}$$

$$\begin{aligned} \langle \chi \mid \psi \rangle &= (- \langle \phi_1 \mid - 2i \langle \phi_2 \mid) (3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle) \\ &= (-1)(3i) \langle \phi_1 \mid \phi_1 \rangle + (-2i)(-7i) \langle \phi_2 \mid \phi_2 \rangle \\ &= -14 - 3i. \end{aligned} \tag{2.44}$$

We see that $\langle \psi \mid \chi \rangle$ is equal to the complex conjugate of $\langle \chi \mid \psi \rangle$.

(c) Let us first calculate $\langle \psi \mid \psi \rangle$ and $\langle \chi \mid \chi \rangle$:

$$\langle \psi \mid \psi \rangle = (-3i \langle \phi_1 \mid + 7i \langle \phi_2 \mid) (3i \mid \phi_1 \rangle - 7i \mid \phi_2 \rangle) = (-3i)(3i) + (7i)(-7i) = 58, \tag{2.45}$$

$$\langle \chi \mid \chi \rangle = (- \langle \phi_1 \mid - 2i \langle \phi_2 \mid) (- \mid \phi_1 \rangle + 2i \mid \phi_2 \rangle) = (-1)(-1) + (-2i)(2i) = 5. \tag{2.46}$$

Since $\langle \psi \mid \chi \rangle = -14 + 3i$ we have $\mid \langle \psi \mid \chi \rangle \mid^2 = 14^2 + 3^2 = 205$. Combining the values of $\mid \langle \psi \mid \chi \rangle \mid^2$, $\langle \psi \mid \psi \rangle$, and $\langle \chi \mid \chi \rangle$, we see that the Schwarz inequality (2.34) is satisfied:

$$205 < (58)(5) \Longrightarrow \mid \langle \psi \mid \chi \rangle \mid^2 < \langle \psi \mid \psi \rangle \langle \chi \mid \chi \rangle. \tag{2.47}$$

(d) First, let us use (2.41) and (2.42) to calculate $\langle \psi + \chi \mid \psi + \chi \rangle$:

$$\begin{aligned} \langle \psi + \chi \mid \psi + \chi \rangle &= [(-1-3i)\langle \phi_1 \mid + 5i \langle \phi_2 \mid][(-1+3i) \mid \phi_1\rangle - 5i \mid \phi_2\rangle] \\ &= (-1-3i)(-1+3i) + (5i)(-5i) \\ &= 35. \end{aligned} \tag{2.48}$$

Since $\langle \psi \mid \psi \rangle = 58$ and $\langle \chi \mid \chi \rangle = 5$, we infer that the triangle inequality (2.36) is satisfied:

$$\sqrt{35} < \sqrt{58} + \sqrt{5} \Longrightarrow \sqrt{\langle \psi + \chi \mid \psi + \chi \rangle} < \sqrt{\langle \psi \mid \psi \rangle} + \sqrt{\langle \chi \mid \chi \rangle}. \tag{2.49}$$

Example 2.5
Consider two states $|\psi_1\rangle = 2i|\phi_1\rangle + |\phi_2\rangle - a|\phi_3\rangle + 4|\phi_4\rangle$ and $|\psi_2\rangle = 3|\phi_1\rangle - i|\phi_2\rangle + 5|\phi_3\rangle - |\phi_4\rangle$, where $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, and $|\phi_4\rangle$ are orthonormal kets, and where a is a constant. Find the value of a so that $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal.

Solution
For the states $|\psi_1\rangle$ and $|\psi_2\rangle$ to be orthogonal, the scalar product $\langle \psi_2 \mid \psi_1 \rangle$ must be zero. Using the relation $\langle \psi_2 \mid = 3\langle \phi_1| + i\langle \phi_2| + 5\langle \phi_3| - \langle \phi_4|$, we can easily find the scalar product

$$\begin{aligned} \langle \psi_2 \mid \psi_1 \rangle &= (3\langle \phi_1| + i\langle \phi_2| + 5\langle \phi_3| - \langle \phi_4|)\,(2i|\phi_1\rangle + |\phi_2\rangle - a|\phi_3\rangle + 4|\phi_4\rangle) \\ &= 7i - 5a - 4. \end{aligned} \tag{2.50}$$

Since $\langle \psi_2 \mid \psi_1 \rangle = 7i - 5a - 4 = 0$, the value of a is $a = (7i-4)/5$.

2.4 Operators

2.4.1 General Definitions

Definition of an operator: An operator[1] $\hat{A}$ is a *mathematical rule* that when applied to a ket $\mid \psi\rangle$ transforms it into another ket $\mid \psi'\rangle$ of the same space and when it acts on a bra $\langle \phi \mid$ transforms it into another bra $\langle \phi' \mid$:

$$\hat{A} \mid \psi\rangle = \mid \psi'\rangle, \qquad \langle \phi \mid \hat{A} = \langle \phi' \mid . \tag{2.51}$$

A similar definition applies to wave functions:

$$\hat{A}\psi(\vec{r}) = \psi'(\vec{r}), \qquad \phi(\vec{r})\hat{A} = \phi'(\vec{r}). \tag{2.52}$$

Examples of operators
Here are some of the operators that we will use in this text:

- Unity operator: it leaves any ket unchanged, $\hat{I} \mid \psi\rangle = \mid \psi\rangle$.
- The gradient operator: $\vec{\nabla}\psi(\vec{r}) = (\partial\psi(\vec{r})/\partial x)\vec{i} + (\partial\psi(\vec{r})/\partial y)\vec{j} + (\partial\psi(\vec{r})/\partial z)\vec{k}$.

[1] The hat on $\hat{A}$ will be used throughout this text to distinguish an operator $\hat{A}$ from a complex number or a matrix A.

- The linear momentum operator: $\vec{P}\psi(\vec{r}) = -i\hbar\vec{\nabla}\psi(\vec{r})$.
- The Laplacian operator: $\nabla^2\psi(\vec{r}) = \partial^2\psi(\vec{r})/\partial x^2 + \partial^2\psi(\vec{r})/\partial y^2 + \partial^2\psi(\vec{r})/\partial z^2$.
- The parity operator: $\hat{\mathcal{P}}\psi(\vec{r}) = \psi(-\vec{r})$.

Products of operators
The product of two operators is generally not commutative:

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}. \tag{2.53}$$

The product of operators is, however, associative:

$$\hat{A}\hat{B}\hat{C} = \hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C}. \tag{2.54}$$

We may also write $\hat{A}^n\hat{A}^m = \hat{A}^{n+m}$. When the product $\hat{A}\hat{B}$ operates on a ket $\mid \psi\rangle$ (the order of application is important), the operator $\hat{B}$ acts first on $\mid \psi\rangle$ and then $\hat{A}$ acts on the new ket $(B \mid \psi\rangle)$:

$$\hat{A}\hat{B} \mid \psi\rangle = \hat{A}(\hat{B} \mid \psi\rangle). \tag{2.55}$$

Similarly, when $\hat{A}\hat{B}\hat{C}\hat{D}$ operates on a ket $\mid \psi\rangle$, $\hat{D}$ acts first, then $\hat{C}$, then $\hat{B}$, and then $\hat{A}$.

When an operator $\hat{A}$ is sandwiched between a bra $\langle\phi \mid$ and a ket $\mid \psi\rangle$, it yields in general a complex number: $\langle\phi \mid \hat{A} \mid \psi\rangle$ = complex number. The quantity $\langle\phi \mid \hat{A} \mid \psi\rangle$ can also be a purely real or a purely imaginary number. **Note:** In evaluating $\langle\phi \mid \hat{A} \mid \psi\rangle$ it does not matter if one first applies $\hat{A}$ to the ket and then takes the bra-ket or one first applies $\hat{A}$ to the bra and then takes the bra-ket; that is $(\langle\phi \mid \hat{A}) \mid \psi\rangle = \langle\phi \mid (\hat{A} \mid \psi\rangle)$.

Linear operators
An operator $\hat{A}$ is said to be *linear* if it obeys the distributive law and, like all operators, it commutes with constants. That is, an operator $\hat{A}$ is linear if, for any vectors $\mid \psi_1\rangle$ and $\mid \psi_2\rangle$ and any complex numbers a_1 and a_2, we have

$$\hat{A}\,(a_1 \mid \psi_1\rangle + a_2 \mid \psi_2\rangle) = a_1\hat{A} \mid \psi_1\rangle + a_2\hat{A} \mid \psi_2\rangle, \tag{2.56}$$

and

$$(\langle\psi_1 \mid a_1 + \langle\psi_2 \mid a_2)\hat{A} = a_1\langle\psi_1 \mid \hat{A} + a_2\langle\psi_2 \mid \hat{A}. \tag{2.57}$$

Remarks

- The *expectation* or *mean* value $\langle\hat{A}\rangle$ of an operator $\hat{A}$ with respect to a state $\mid \psi\rangle$ is defined by

$$\langle\hat{A}\rangle = \frac{\langle\psi \mid \hat{A} \mid \psi\rangle}{\langle\psi \mid \psi\rangle}. \tag{2.58}$$

- The quantity $\mid \phi\rangle\langle\psi \mid$ (i.e., the product of a ket with a bra) is a linear operator in Dirac's notation. To see this, when $\mid \phi\rangle\langle\psi \mid$ is applied to a ket $\mid \psi'\rangle$, we obtain another ket:

$$\mid \phi\rangle\langle\psi \mid \psi'\rangle = \langle\psi \mid \psi'\rangle \mid \phi\rangle, \tag{2.59}$$

  since $\langle\psi \mid \psi'\rangle$ is a complex number.

- Products of the type $\mid \psi\rangle\hat{A}$ and $\hat{A}\langle\psi \mid$ (i.e., when an operator stands on the right of a ket or on the left of a bra) are forbidden. They are not operators, or kets, or bras; they have no mathematical or physical meanings (see equation (2.219) for an illustration).

2.4.2 Hermitian Adjoint

The Hermitian adjoint or conjugate[2], $\alpha^\dagger$, of a complex number α is the complex conjugate of this number: $\alpha^\dagger = \alpha^*$. The Hermitian adjoint, or simply the adjoint, $\hat{A}^\dagger$, of an operator $\hat{A}$ is defined by this relation:

$$\langle \psi \mid \hat{A}^\dagger \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*. \tag{2.60}$$

Properties of the Hermitian conjugate rule
To obtain the Hermitian adjoint of any expression, we must cyclically reverse the order of the factors and make three replacements:

- Replace constants by their complex conjugates: $\alpha^\dagger = \alpha^*$.
- Replace kets (bras) by the corresponding bras (kets): $(\mid \psi \rangle)^\dagger = \langle \psi \mid$ and $(\langle \psi \mid)^\dagger = \mid \psi \rangle$.
- Replace operators by their adjoints.

Following these rules, we can write

$$\begin{aligned}
(\hat{A}^\dagger)^\dagger &= \hat{A}, && (2.61)\\
(a\hat{A})^\dagger &= a^* \hat{A}^\dagger, && (2.62)\\
(\hat{A}^n)^\dagger &= (\hat{A}^\dagger)^n, && (2.63)\\
(\hat{A} + \hat{B} + \hat{C} + \hat{D})^\dagger &= \hat{A}^\dagger + \hat{B}^\dagger + \hat{C}^\dagger + \hat{D}^\dagger, && (2.64)\\
(\hat{A}\hat{B}\hat{C}\hat{D})^\dagger &= \hat{D}^\dagger \hat{C}^\dagger \hat{B}^\dagger \hat{A}^\dagger, && (2.65)\\
(\hat{A}\hat{B}\hat{C}\hat{D} \mid \psi\rangle)^\dagger &= \langle \psi \mid D^\dagger C^\dagger B^\dagger A^\dagger. && (2.66)
\end{aligned}$$

The Hermitian adjoint of the operator $\mid \psi\rangle\langle\phi \mid$ is given by

$$(\mid \psi\rangle\langle\phi \mid)^\dagger = \mid \phi\rangle\langle\psi \mid . \tag{2.67}$$

Operators act inside kets and bras, respectively, as follows:

$$\mid \alpha \hat{A}\psi\rangle = \alpha \hat{A} \mid \psi\rangle, \qquad \langle \alpha \hat{A}\psi \mid = \alpha^* \langle \psi \mid \hat{A}^\dagger. \tag{2.68}$$

Note also that $\langle \alpha \hat{A}^\dagger \psi \mid = \alpha^* \langle \psi \mid (\hat{A}^\dagger)^\dagger = \alpha^* \langle \psi \mid \hat{A}$. Hence, we can also write:

$$\langle \psi \mid \hat{A} \mid \phi \rangle = \langle \hat{A}^\dagger \psi \mid \phi \rangle = \langle \psi \mid \hat{A}\phi \rangle. \tag{2.69}$$

Hermitian and skew-Hermitian operators
An operator $\hat{A}$ is said to be *Hermitian* if it is equal to its adjoint $\hat{A}^\dagger$:

$$\boxed{\hat{A} = \hat{A}^\dagger \quad \text{or} \quad \langle \psi \mid \hat{A} \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*.} \tag{2.70}$$

[2]The terms "adjoint" and "conjugate" are used indiscriminately.

On the other hand, an operator $\hat{B}$ is said to be *skew-Hermitian* or *anti-Hermitian* if

$$\boxed{\hat{B}^\dagger = -\hat{B} \qquad \text{or} \quad \langle\psi \mid \hat{B} \mid \phi\rangle = -\langle\phi \mid \hat{B} \mid \psi\rangle^*.} \tag{2.71}$$

Remark

The Hermitian adjoint of an operator is not, in general, equal to its complex conjugate: $\hat{A}^\dagger \neq \hat{A}^*$.

Example 2.6

(a) Discuss the hermiticity of the operators $(\hat{A} + \hat{A}^\dagger)$, $i(\hat{A} + \hat{A}^\dagger)$, and $i(\hat{A} - \hat{A}^\dagger)$.

(b) Find the Hermitian adjoint of $f(\hat{A}) = (1 + i\hat{A} + 3\hat{A}^2)(1 - 2i\hat{A} - 9\hat{A}^2)/(5 + 7\hat{A})$.

(c) Show that the expectation value of a Hermitian operator is real and that of an anti-Hermitian operator is imaginary.

Solution

(a) The operator $\hat{B} = \hat{A} + \hat{A}^\dagger$ is Hermitian regardless of whether or not $\hat{A}$ is Hermitian, since

$$\hat{B}^\dagger = (\hat{A} + \hat{A}^\dagger)^\dagger = \hat{A}^\dagger + \hat{A} = \hat{B}. \tag{2.72}$$

Similarly, the operator $i(\hat{A} - \hat{A}^\dagger)$ is also Hermitian; but $i(\hat{A} + \hat{A}^\dagger)$ is anti-Hermitian, since $[i(\hat{A} + \hat{A}^\dagger)]^\dagger = -i(\hat{A} + \hat{A}^\dagger)$.

(b) Since the Hermitian adjoint of an operator function $f(\hat{A})$ is given by $f^\dagger(\hat{A}) = f^*(\hat{A}^\dagger)$, we can write

$$\left(\frac{(1 + i\hat{A} + 3\hat{A}^2)(1 - 2i\hat{A} - 9\hat{A}^2)}{5 + 7\hat{A}}\right)^\dagger = \frac{(1 + 2i\hat{A}^\dagger - 9\hat{A}^{\dagger 2})(1 - i\hat{A}^\dagger + 3\hat{A}^{\dagger 2})}{5 + 7\hat{A}^\dagger}. \tag{2.73}$$

(c) From (2.70) we immediately infer that the expectation value of a Hermitian operator is real, for it satisfies the following property:

$$\langle\psi \mid \hat{A} \mid \psi\rangle = \langle\psi \mid \hat{A} \mid \psi\rangle^*; \tag{2.74}$$

that is, if $\hat{A}^\dagger = \hat{A}$ then $\langle\psi \mid \hat{A} \mid \psi\rangle$ is real. Similarly, for an anti-Hermitian operator, $\hat{B}^\dagger = -\hat{B}$, we have

$$\langle\psi \mid \hat{B} \mid \psi\rangle = -\langle\psi \mid \hat{B} \mid \psi\rangle^*, \tag{2.75}$$

which means that $\langle\psi \mid \hat{B} \mid \psi\rangle$ is a purely imaginary number.

2.4.3 Projection Operators

An operator $\hat{P}$ is said to be a *projection operator* if it is Hermitian and equal to its own square:

$$\hat{P}^\dagger = \hat{P}, \qquad \hat{P}^2 = \hat{P}. \tag{2.76}$$

The unit operator $\hat{I}$ is a simple example of a projection operator, since $\hat{I}^\dagger = \hat{I}, \quad \hat{I}^2 = \hat{I}$.

Properties of projection operators

- The product of two commuting projection operators, $\hat{P}_1$ and $\hat{P}_2$, is also a projection operator, since

$$(\hat{P}_1\hat{P}_2)^\dagger = \hat{P}_2^\dagger\hat{P}_1^\dagger = \hat{P}_2\hat{P}_1 = \hat{P}_1\hat{P}_2 \text{ and } (\hat{P}_1\hat{P}_2)^2 = \hat{P}_1\hat{P}_2\hat{P}_1\hat{P}_2 = \hat{P}_1^2\hat{P}_2^2 = \hat{P}_1\hat{P}_2. \tag{2.77}$$

- The sum of two projection operators is generally not a projection operator.
- Two projection operators are said to be orthogonal if their product is zero.
- For a sum of projection operators $\hat{P}_1 + \hat{P}_2 + \hat{P}_3 + \cdots$ to be a projection operator, it is necessary and sufficient that these projection operators be mutually orthogonal (i.e., the cross-product terms must vanish).

Example 2.7
Show that the operator $|\ \psi\rangle\langle\psi\ |$ is a projection operator only when $|\ \psi\rangle$ is normalized.

Solution
It is easy to ascertain that the operator $|\ \psi\rangle\langle\psi\ |$ is Hermitian, since $(|\ \psi\rangle\langle\psi\ |)^\dagger = |\ \psi\rangle\langle\psi\ |$. As for the square of this operator, it is given by

$$(|\ \psi\rangle\langle\psi\ |)^2 = (|\ \psi\rangle\langle\psi\ |)(|\ \psi\rangle\langle\psi\ |) = |\ \psi\rangle\langle\psi\ |\ \psi\rangle\langle\psi\ |\ . \tag{2.78}$$

Thus, if $|\ \psi\rangle$ is normalized, we have $(|\ \psi\rangle\langle\psi\ |)^2 = |\ \psi\rangle\langle\psi\ |$. In sum, if the state $|\ \psi\rangle$ is normalized, the product of the ket $|\ \psi\rangle$ with the bra $\langle\psi\ |$ is a projection operator.

2.4.4 Commutator Algebra

The *commutator* of two operators $\hat{A}$ and $\hat{B}$, denoted by $[\hat{A},\ \hat{B}]$, is defined by

$$\boxed{[\hat{A},\ \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},} \tag{2.79}$$

and the *anticommutator* $\{\hat{A},\ \hat{B}\}$ is defined by

$$\boxed{\{\hat{A},\ \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.} \tag{2.80}$$

Two operators are said to commute if their commutator is equal to zero and hence $\hat{A}\hat{B} = \hat{B}\hat{A}$. Any operator commutes with itself:

$$[\hat{A},\ \hat{A}] = 0. \tag{2.81}$$

Note that if two operators are Hermitian and their product is also Hermitian, these operators commute:

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger = \hat{B}\hat{A}, \tag{2.82}$$

and since $(\hat{A}\hat{B})^\dagger = \hat{A}\hat{B}$ we have $\hat{A}\hat{B} = \hat{B}\hat{A}$.

As an example, we may mention the commutators involving the x-position operator, $\hat{X}$, and the x-component of the momentum operator, $\hat{P}_x = -i\hbar\partial/\partial x$, as well as the y and the z components

$$\boxed{[\hat{X},\ \hat{P}_x] = i\hbar\hat{I}, \qquad [\hat{Y},\ \hat{P}_y] = i\hbar\hat{I}, \qquad [\hat{Z},\ \hat{P}_z] = i\hbar\hat{I},} \tag{2.83}$$

where $\hat{I}$ is the unit operator.

Properties of commutators
Using the commutator relation (2.79), we can establish the following properties:

- Antisymmetry:
$$[\hat{A},\ \hat{B}] = -[\hat{B},\ \hat{A}] \tag{2.84}$$

- Linearity:
$$[\hat{A},\ \hat{B} + \hat{C} + \hat{D} + \cdots] = [\hat{A},\ \hat{B}] + [\hat{A},\ \hat{C}] + [\hat{A},\ \hat{D}] + \cdots \tag{2.85}$$

- Hermitian conjugate of a commutator:
$$[\hat{A},\ \hat{B}]^\dagger = [\hat{B}^\dagger,\ \hat{A}^\dagger] \tag{2.86}$$

- Distributivity:
$$[\hat{A},\ \hat{B}\hat{C}] = [\hat{A},\ \hat{B}]\hat{C} + \hat{B}[\hat{A},\ \hat{C}] \tag{2.87}$$
$$[\hat{A}\hat{B},\ \hat{C}] = \hat{A}[\hat{B},\ \hat{C}] + [\hat{A},\ \hat{C}]\hat{B} \tag{2.88}$$

- Jacobi identity:
$$[\hat{A}, [\hat{B},\ \hat{C}]] + [\hat{B},\ [\hat{C},\ \hat{A}]] + [\hat{C},\ [\hat{A},\ \hat{B}]] = 0 \tag{2.89}$$

- By repeated applications of (2.87), we can show that
$$[\hat{A},\ \hat{B}^n] = \sum_{j=0}^{n-1} \hat{B}^j [\hat{A},\ \hat{B}] \hat{B}^{n-j-1} \tag{2.90}$$
$$[\hat{A}^n,\ \hat{B}] = \sum_{j=0}^{n-1} \hat{A}^{n-j-1} [\hat{A},\ \hat{B}] \hat{A}^j \tag{2.91}$$

- Operators commute with scalars: an operator $\hat{A}$ commutes with any scalar b:
$$[\hat{A},\ b] = 0 \tag{2.92}$$

Example 2.8
(a) Show that the commutator of two Hermitian operators is anti-Hermitian.
(b) Evaluate the commutator $[\hat{A},\ [\hat{B},\ \hat{C}]\hat{D}]$.

Solution

(a) If $\hat{A}$ and $\hat{B}$ are Hermitian, we can write

$$[\hat{A},\ \hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}^\dagger\hat{A}^\dagger - \hat{A}^\dagger\hat{B}^\dagger = \hat{B}\hat{A} - \hat{A}\hat{B} = -[\hat{A},\ \hat{B}]; \tag{2.93}$$

that is, the commutator of $\hat{A}$ and $\hat{B}$ is anti-Hermitian: $[\hat{A},\ \hat{B}]^\dagger = -[\hat{A},\ \hat{B}]$.

(b) Using the distributivity relation (2.87), we have

$$\begin{aligned}[\hat{A},\ [\hat{B},\ \hat{C}]\hat{D}] &= [\hat{B},\ \hat{C}][\hat{A},\ \hat{D}] + [\hat{A},\ [\hat{B},\ \hat{C}]]\hat{D} \\ &= (\hat{B}\hat{C} - \hat{C}\hat{B})(\hat{A}\hat{D} - \hat{D}\hat{A}) + \hat{A}(\hat{B}\hat{C} - \hat{C}\hat{B})\hat{D} - (\hat{B}\hat{C} - \hat{C}\hat{B})\hat{A}\hat{D} \\ &= \hat{C}\hat{B}\hat{D}\hat{A} - \hat{B}\hat{C}\hat{D}\hat{A} + \hat{A}\hat{B}\hat{C}\hat{D} - \hat{A}\hat{C}\hat{B}\hat{D}. \end{aligned} \tag{2.94}$$

2.4.5 Uncertainty Relation between Two Operators

An interesting application of the commutator algebra is to derive a general relation giving the uncertainties product of two operators, $\hat{A}$ and $\hat{B}$. In particular, we want to give a formal derivation of Heisenberg's uncertainty relations.

Let $\langle\hat{A}\rangle$ and $\langle\hat{B}\rangle$ denote the expectation values of two Hermitian operators $\hat{A}$ and $\hat{B}$ with respect to a normalized state vector $|\ \psi\rangle$: $\langle\hat{A}\rangle = \langle\psi\ |\ \hat{A}\ |\ \psi\rangle$ and $\langle\hat{B}\rangle = \langle\psi\ |\ \hat{B}\ |\ \psi\rangle$. Introducing the operators $\Delta\hat{A}$ and $\Delta\hat{B}$,

$$\Delta\hat{A} = \hat{A} - \langle\hat{A}\rangle, \qquad \Delta\hat{B} = \hat{B} - \langle\hat{B}\rangle, \tag{2.95}$$

we have $(\Delta\hat{A})^2 = \hat{A}^2 - 2\hat{A}\langle\hat{A}\rangle + \langle\hat{A}\rangle^2$ and $(\Delta\hat{B})^2 = \hat{B}^2 - 2\hat{B}\langle\hat{B}\rangle + \langle\hat{B}\rangle^2$, and hence

$$\langle\psi\ |\ (\Delta\hat{A})^2\ |\ \psi\rangle = \langle(\Delta\hat{A})^2\rangle = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2, \qquad \langle(\Delta\hat{B})^2\rangle = \langle\hat{B}^2\rangle - \langle\hat{B}\rangle^2, \tag{2.96}$$

where $\langle\hat{A}^2\rangle = \langle\psi\ |\ \hat{A}^2\ |\ \psi\rangle$ and $\langle\hat{B}^2\rangle = \langle\psi\ |\ \hat{B}^2\ |\ \psi\rangle$. The *uncertainties* ΔA and ΔB are defined by

$$\boxed{\Delta A = \sqrt{\langle(\Delta\hat{A})^2\rangle} = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2}, \qquad \Delta B = \sqrt{\langle(\Delta\hat{B})^2\rangle} = \sqrt{\langle\hat{B}^2\rangle - \langle\hat{B}\rangle^2}.} \tag{2.97}$$

Let us write the action of the operators (2.95) on any state $|\ \psi\rangle$ as follows:

$$|\ \chi\rangle = \Delta\hat{A}\ |\ \psi\rangle = \left(\hat{A} - \langle\hat{A}\rangle\right)|\ \psi\rangle, \qquad |\ \phi\rangle = \Delta\hat{B}\ |\ \psi\rangle = \left(\hat{B} - \langle\hat{B}\rangle\right)|\ \psi\rangle. \tag{2.98}$$

The Schwarz inequality for the states $|\ \chi\rangle$ and $|\ \phi\rangle$ is given by

$$\langle\chi\ |\ \chi\rangle\langle\phi\ |\ \phi\rangle \geq |\langle\chi\ |\ \phi\rangle|^2. \tag{2.99}$$

Since $\hat{A}$ and $\hat{B}$ are Hermitian, $\Delta\hat{A}$ and $\Delta\hat{B}$ must also be Hermitian: $\Delta\hat{A}^\dagger = \hat{A}^\dagger - \langle\hat{A}\rangle = \hat{A} - \langle\hat{A}\rangle = \Delta\hat{A}$ and $\Delta\hat{B}^\dagger = \hat{B} - \langle\hat{B}\rangle = \Delta\hat{B}$. Thus, we can show the following three relations:

$$\langle\chi\ |\ \chi\rangle = \langle\psi\ |\ (\Delta\hat{A})^2\ |\ \psi\rangle, \quad \langle\phi\ |\ \phi\rangle = \langle\psi\ |\ (\Delta\hat{B})^2\ |\ \psi\rangle, \quad \langle\chi\ |\ \phi\rangle = \langle\psi\ |\ \Delta\hat{A}\Delta\hat{B}\ |\ \psi\rangle. \tag{2.100}$$

For instance, since $\Delta\hat{A}^\dagger = \Delta\hat{A}$ we have $\langle\chi \mid \chi\rangle = \langle\psi \mid \Delta\hat{A}^\dagger\Delta\hat{A} \mid \psi\rangle = \langle\psi \mid (\Delta\hat{A})^2 \mid \psi\rangle = \langle(\Delta\hat{A})^2\rangle$. Hence, the Schwarz inequality (2.99) becomes

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \;\geq\; \left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2. \tag{2.101}$$

Notice that the last term $\Delta\hat{A}\Delta\hat{B}$ of this equation can be written as

$$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}[\Delta\hat{A},\ \Delta\hat{B}] + \frac{1}{2}\{\Delta\hat{A},\ \Delta\hat{B}\} = \frac{1}{2}[\hat{A},\ \hat{B}] + \frac{1}{2}\{\Delta\hat{A},\ \Delta\hat{B}\}, \tag{2.102}$$

where we have used the fact that $[\Delta\hat{A},\ \Delta\hat{B}] = [\hat{A},\ \hat{B}]$. Since $[\hat{A},\ \hat{B}]$ is anti-Hermitian and $\{\Delta\hat{A},\ \Delta\hat{B}\}$ is Hermitian and since the expectation value of a Hermitian operator is real and that the expectation value of an anti-Hermitian operator is imaginary (see Example 2.6), the expectation value $\langle\Delta\hat{A}\Delta\hat{B}\rangle$ of (2.102) becomes equal to the sum of a real part $\langle\{\Delta\hat{A},\ \Delta\hat{B}\}\rangle/2$ and an imaginary part $\langle[\hat{A},\ \hat{B}]\rangle/2$; hence

$$\left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 = \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2 + \frac{1}{4}\left|\langle\{\Delta\hat{A},\ \Delta\hat{B}\}\rangle\right|^2. \tag{2.103}$$

Since the last term is a positive real number, we can infer the following relation:

$$\left|\langle\Delta\hat{A}\Delta\hat{B}\rangle\right|^2 \;\geq\; \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2. \tag{2.104}$$

Comparing equations (2.101) and (2.104), we conclude that

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \;\geq\; \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2, \tag{2.105}$$

which (by taking its square root) can be reduced to

$$\boxed{\Delta A\Delta B \geq \frac{1}{2}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|.} \tag{2.106}$$

This uncertainty relation plays an important role in the formalism of quantum mechanics. Its application to position and momentum operators leads to the Heisenberg uncertainty relations, which represent one of the cornerstones of quantum mechanics; see the next example.

Example 2.9 (Heisenberg uncertainty relations)
Find the uncertainty relations between the components of the position and the momentum operators.

Solution
By applying (2.106) to the x-components of the position operator $\hat{X}$, and the momentum operator $\hat{P}_x$, we obtain $\Delta x\,\Delta p_x \;\geq\; \frac{1}{2} \mid \langle[\hat{X},\ \hat{P}_x]\rangle \mid$. But since $[\hat{X},\ \hat{P}_x] = i\hbar\hat{I}$, we have $\Delta x\,\Delta p_x \geq \hbar/2$; the uncertainty relations for the $y-$ and $z-$ components follow immediately:

$$\boxed{\Delta x\,\Delta p_x \geq \frac{\hbar}{2}, \qquad \Delta y\,\Delta p_y \geq \frac{\hbar}{2}, \qquad \Delta z\,\Delta p_z \geq \frac{\hbar}{2}.} \tag{2.107}$$

These are the Heisenberg uncertainty relations.

2.4.6 Functions of Operators

Let $F(\hat{A})$ be a function of an operator $\hat{A}$. If $\hat{A}$ is a linear operator, we can Taylor expand $F(\hat{A})$ in a power series of $\hat{A}$:

$$F(\hat{A}) = \sum_{n=0}^{\infty} a_n \hat{A}^n, \tag{2.108}$$

where a_n is just an expansion coefficient. As an illustration of an operator function, consider $e^{a\hat{A}}$, where a is a scalar which can be complex or real. We can expand it as follows:

$$e^{a\hat{A}} = \sum_{n=0}^{\infty} \frac{a^n}{n!} \hat{A}^n = \hat{I} + a\hat{A} + \frac{a^2}{2!}\hat{A}^2 + \frac{a^3}{3!}\hat{A}^3 + \cdots. \tag{2.109}$$

Commutators involving function operators
If $\hat{A}$ commutes with another operator $\hat{B}$, then $\hat{B}$ commutes with any operator function that depends on $\hat{A}$:

$$[\hat{A},\ \hat{B}] = 0 \quad \Longrightarrow \quad [\hat{B},\ F(\hat{A})] = 0; \tag{2.110}$$

in particular, $F(\hat{A})$ commutes with $\hat{A}$ and with any other function, $G(\hat{A})$, of $\hat{A}$:

$$[\hat{A},\ F(\hat{A})] = 0, \qquad [\hat{A}^n,\ F(\hat{A})] = 0, \qquad [F(\hat{A}),\ G(\hat{A})] = 0. \tag{2.111}$$

Hermitian adjoint of function operators
The adjoint of $F(\hat{A})$ is given by

$$[F(\hat{A})]^\dagger = F^*(\hat{A}^\dagger). \tag{2.112}$$

Note that if $\hat{A}$ is Hermitian, $F(\hat{A})$ is not necessarily Hermitian; $F(\hat{A})$ will be Hermitian only if F is a real function and $\hat{A}$ is Hermitian. An example is

$$(e^{\hat{A}})^\dagger = e^{\hat{A}^\dagger}, \qquad (e^{i\hat{A}})^\dagger = e^{-i\hat{A}^\dagger}, \qquad (e^{i\alpha\hat{A}})^\dagger = e^{-i\alpha^*\hat{A}^\dagger}, \tag{2.113}$$

where α is a complex number. So if $\hat{A}$ is Hermitian, an operator function which can be expanded as $F(\hat{A}) = \sum_{n=0}^{\infty} a_n \hat{A}^n$ will be Hermitian only if the expansion coefficients a_n are real numbers. But in general, $F(\hat{A})$ is not Hermitian even if $\hat{A}$ is Hermitian, since

$$F^*(\hat{A}^\dagger) = \sum_{n=0}^{\infty} a_n^* (\hat{A}^\dagger)^n. \tag{2.114}$$

Relations involving function operators
Note that

$$[\hat{A},\ \hat{B}] \neq 0 \quad \Longrightarrow \quad [\hat{B},\ F(\hat{A})] \neq 0; \tag{2.115}$$

in particular, $e^{\hat{A}}e^{\hat{B}} \neq e^{\hat{A}+\hat{B}}$. Using (2.109) we can ascertain that

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}e^{[\hat{A},\ \hat{B}]/2}, \tag{2.116}$$

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A},\ \hat{B}] + \frac{1}{2!}[\hat{A},\ [\hat{A},\ \hat{B}]] + \frac{1}{3!}[\hat{A},\ [\hat{A},\ [\hat{A},\ \hat{B}]]] + \cdots. \tag{2.117}$$

2.4.7 Inverse and Unitary Operators

Inverse of an operator: Assuming it exists[3] the *inverse* $\hat{A}^{-1}$ of a linear operator $\hat{A}$ is defined by the relation

$$\hat{A}^{-1}\hat{A} = \hat{A}\hat{A}^{-1} = \hat{I}, \tag{2.118}$$

where $\hat{I}$ is the unit operator, the operator that leaves any state $|\psi\rangle$ unchanged.

Quotient of two operators: Dividing an operator $\hat{A}$ by another operator $\hat{B}$ (provided that the inverse $\hat{B}^{-1}$ exists) is equivalent to multiplying $\hat{A}$ by $\hat{B}^{-1}$:

$$\frac{\hat{A}}{\hat{B}} = \hat{A}\hat{B}^{-1}. \tag{2.119}$$

The side on which the quotient is taken matters:

$$\frac{\hat{A}}{\hat{B}} = \hat{A}\frac{\hat{I}}{\hat{B}} = \hat{A}\hat{B}^{-1} \quad \text{and} \quad \frac{\hat{I}}{\hat{B}}\hat{A} = \hat{B}^{-1}\hat{A}. \tag{2.120}$$

In general, we have $\hat{A}\hat{B}^{-1} \neq \hat{B}^{-1}\hat{A}$. For an illustration of these ideas, see Problem 2.12. We may mention here the following properties about the inverse of operators:

$$\left(\hat{A}\hat{B}\hat{C}\hat{D}\right)^{-1} = \hat{D}^{-1}\hat{C}^{-1}\hat{B}^{-1}\hat{A}^{-1}, \qquad \left(\hat{A}^{n}\right)^{-1} = \left(\hat{A}^{-1}\right)^{n}. \tag{2.121}$$

Unitary operators: A linear operator $\hat{U}$ is said to be *unitary* if its inverse $\hat{U}^{-1}$ is equal to its adjoint $\hat{U}^{\dagger}$:

$$\hat{U}^{\dagger} = \hat{U}^{-1} \qquad \text{or} \qquad \hat{U}\hat{U}^{\dagger} = \hat{U}^{\dagger}\hat{U} = \hat{I}. \tag{2.122}$$

The product of two unitary operators is also unitary, since

$$(\hat{U}\hat{V})(\hat{U}\hat{V})^{\dagger} = (\hat{U}\hat{V})(\hat{V}^{\dagger}\hat{U}^{\dagger}) = \hat{U}(\hat{V}\hat{V}^{\dagger})\hat{U}^{\dagger} = \hat{U}\hat{U}^{\dagger} = \hat{I}, \tag{2.123}$$

or $(\hat{U}\hat{V})^{\dagger} = (\hat{U}\hat{V})^{-1}$. This result can be generalized to any number of operators; the product of a number of unitary operators is also unitary, since

$$\begin{aligned}(\hat{A}\hat{B}\hat{C}\hat{D}\cdots)(\hat{A}\hat{B}\hat{C}\hat{D}\cdots)^{\dagger} &= \hat{A}\hat{B}\hat{C}\hat{D}(\cdots)\hat{D}^{\dagger}\hat{C}^{\dagger}\hat{B}^{\dagger}\hat{A}^{\dagger} = \hat{A}\hat{B}\hat{C}(\hat{D}\hat{D}^{\dagger})\hat{C}^{\dagger}\hat{B}^{\dagger}\hat{A}^{\dagger} \\ &= \hat{A}\hat{B}(\hat{C}\hat{C}^{\dagger})\hat{B}^{\dagger}\hat{A}^{\dagger} = \hat{A}(\hat{B}\hat{B}^{\dagger})\hat{A}^{\dagger} \\ &= \hat{A}\hat{A}^{\dagger} = \hat{I}, \end{aligned} \tag{2.124}$$

or $(\hat{A}\hat{B}\hat{C}\hat{D}\cdots)^{\dagger} = (\hat{A}\hat{B}\hat{C}\hat{D}\cdots)^{-1}$.

Example 2.10 (Unitary operator)
What conditions must the parameter ε and the operator $\hat{G}$ satisfy so that the operator $\hat{U} = e^{i\varepsilon\hat{G}}$ is unitary?

[3] Not every operator has an inverse, just as in the case of matrices. The inverse of a matrix exists only when its determinant is nonzero.